

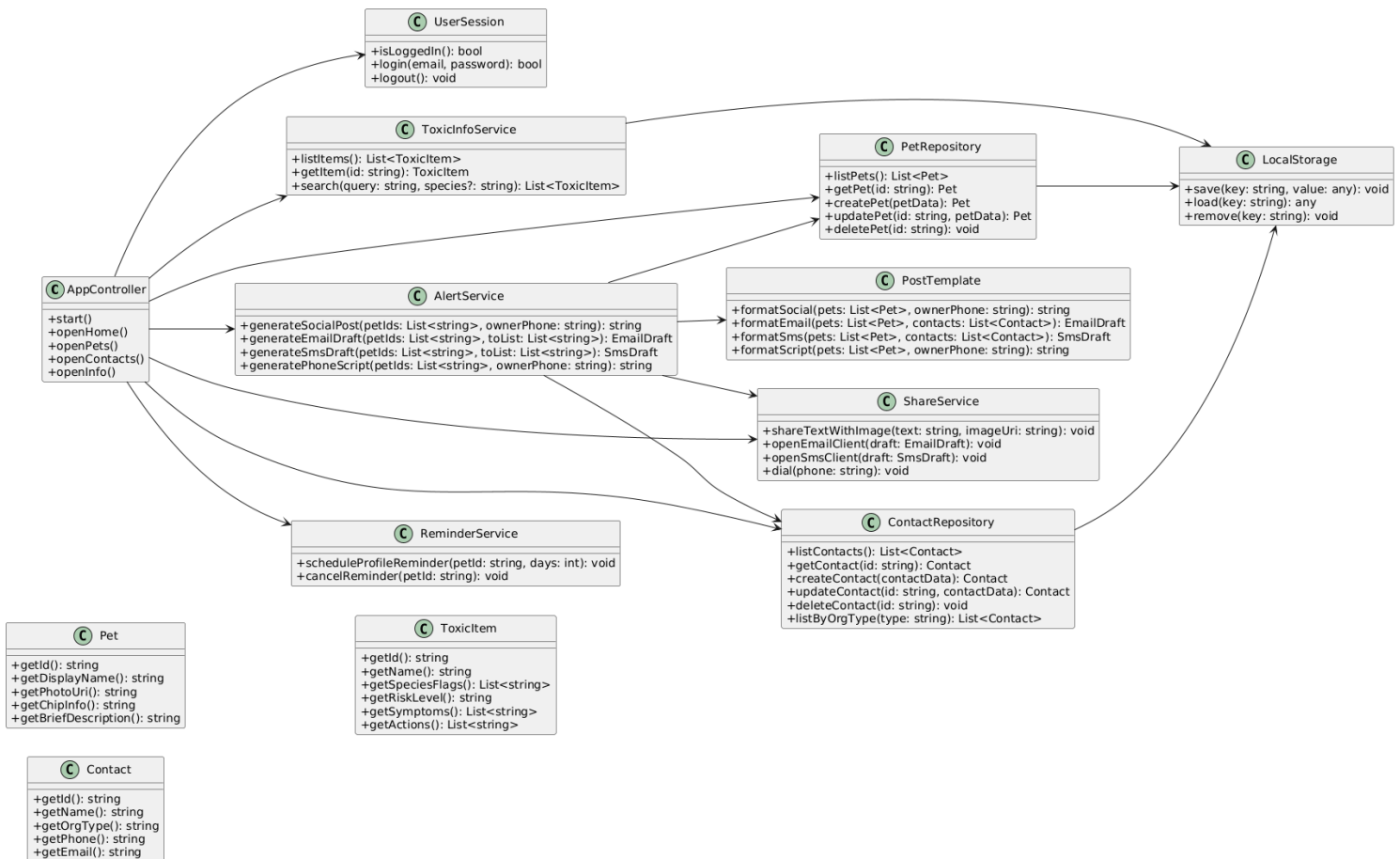
Product Design

Team <Sunrise Inc>

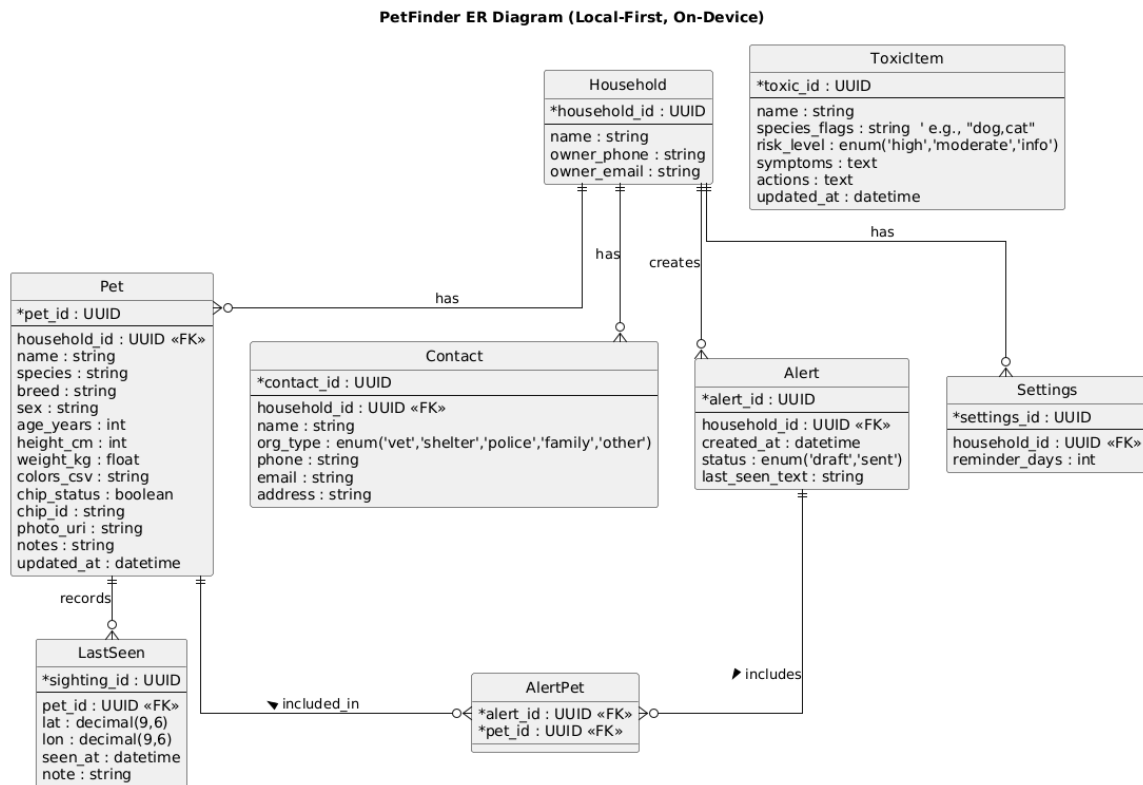
Anjali Fernando, Tanvi Biswal,
Kimberley Juarez, Ojaswi
Subedi, Bibek Pandey

<i>Revision Number</i>	<i>Revision Date</i>	<i>Summary of Changes</i>	<i>Author(s)</i>
0.1	10/08/2025	Initial design document information added	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.2	11/08/2025	Screenshot added from the actual implementation in android emulator. New design summary and design rationale added.	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.3	12/02/2025	Updated UI screenshots, added Login/Signup designs, updated Information Architecture to include Firebase Authentication, and added the Missing Alerts module	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey
0.4	02/14/2026	Migrated pet data to Firebase Firestore & Storage, added microchip field, updated Information Architecture for cloud synchronization	Anjali Fernando, Tanvi Biswal, Kimberley Juarez, Ojaswi Subedi, Bibek Pandey

Class Diagram(s)



ER Diagram(s)

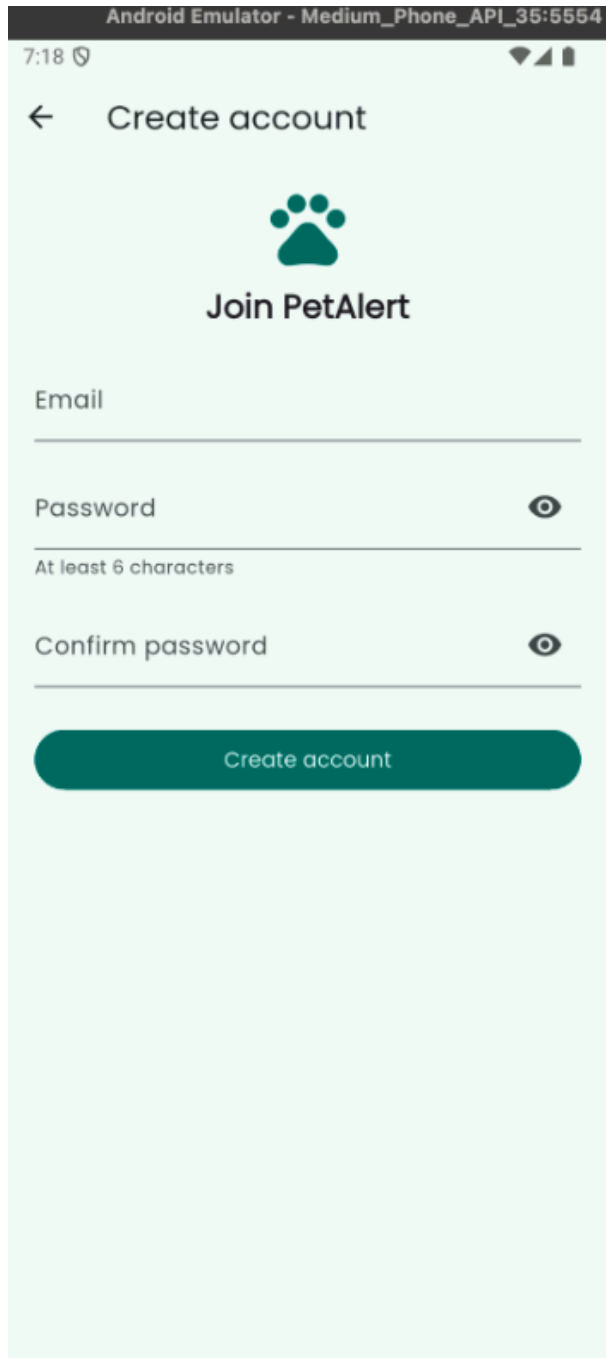


Information Architecture Diagram

PetAlert originally followed a fully local-first architecture in Sprint 2. In Sprint 3, the architecture was expanded to include Firebase Authentication. In Sprint 4, the system was migrated to a cloud-based architecture using Firebase Firestore and Firebase Storage. Pet profiles, images, and microchip information are now securely stored in Firestore and synchronized across devices, improving persistence and scalability.

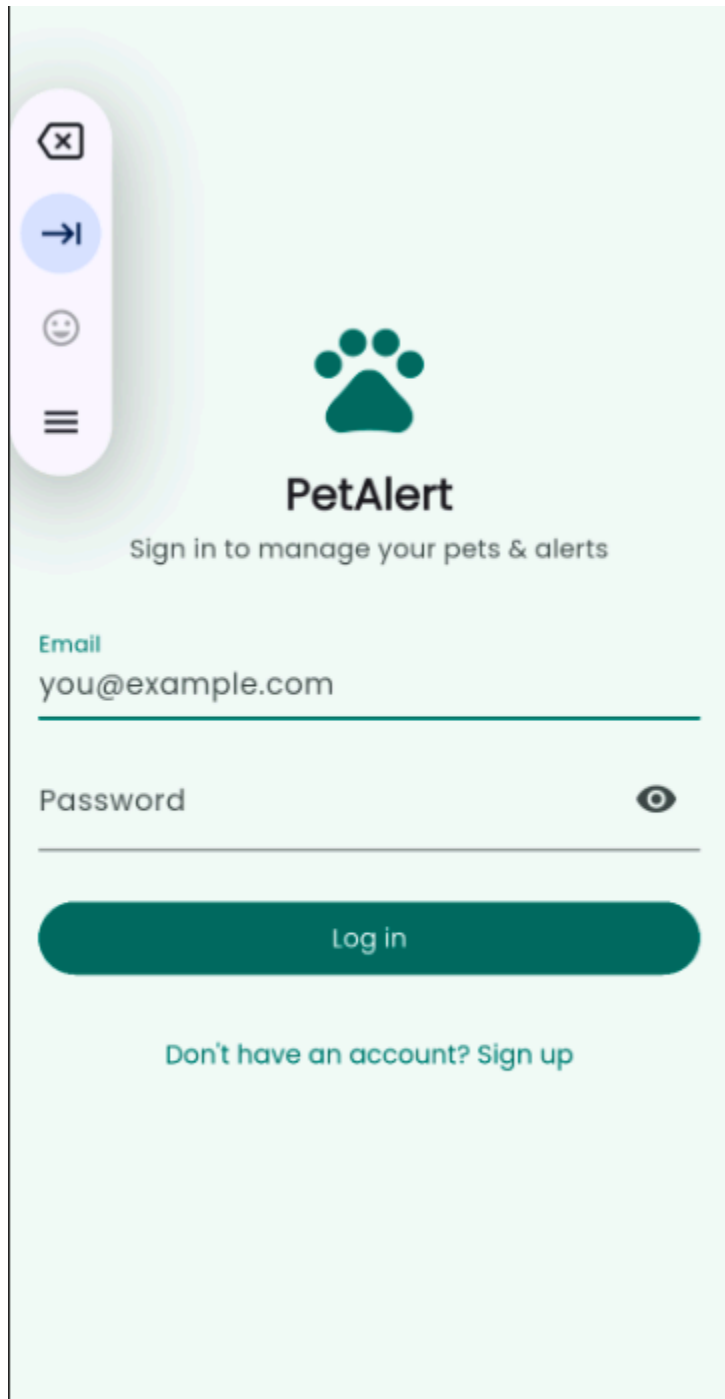
Screenshot(s)

Added signup



The screenshot shows the 'Create account' screen of the PetAlert app. At the top, there is a back arrow and the text 'Create account'. Below this is a green paw print icon and the text 'Join PetAlert'. The form consists of three input fields: 'Email', 'Password', and 'Confirm password'. The 'Password' field has a toggle icon (an eye) to its right, and a note 'At least 6 characters' is displayed below it. The 'Confirm password' field also has a toggle icon to its right. At the bottom of the form is a large green button with the text 'Create account'.

Added Login



A mobile app login screen for 'PetAlert'. The background is a light mint green. On the left, there is a vertical white pill-shaped menu with four icons: a house with an 'x', a blue circle with a right arrow, a smiley face, and a hamburger menu. In the center, there is a dark green paw print icon above the text 'PetAlert' in a bold, dark green font. Below the name is the text 'Sign in to manage your pets & alerts' in a smaller, grey font. The login form consists of two fields: 'Email' with the placeholder 'you@example.com' and 'Password' with a toggle eye icon. Below the fields is a large, rounded dark green button with the text 'Log in' in white. At the bottom, there is a link that says 'Don't have an account? Sign up' in a teal color.







PetAlert
Sign in to manage your pets & alerts

Email
you@example.com

Password 

Log in

Don't have an account? Sign up

Creating Missing Alert

 Create Missing Alert

Select pet

pupp

Last seen location

Last seen date & time

Not set



Contact name (optional)

Contact phone (optional)

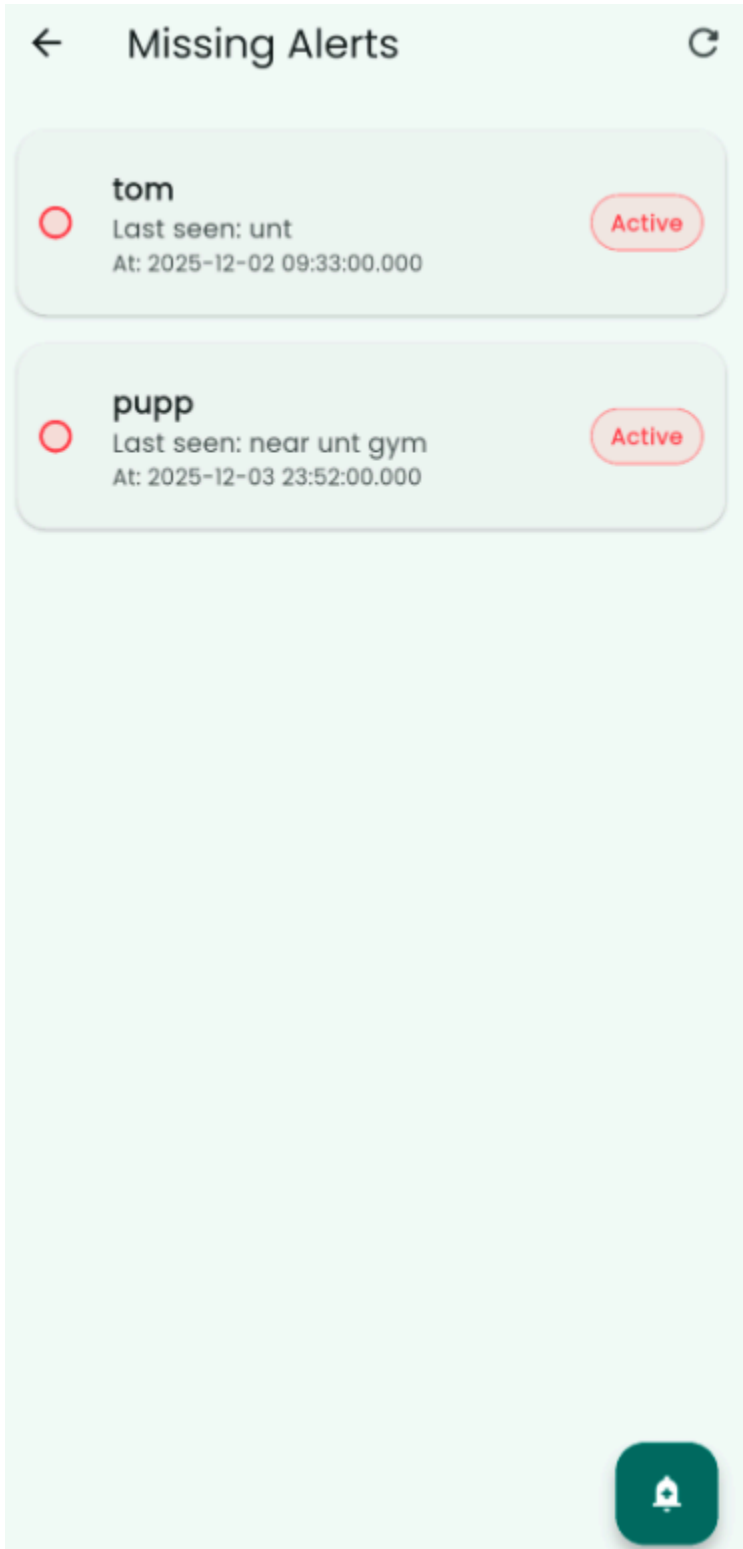
Notes (optional)

Status

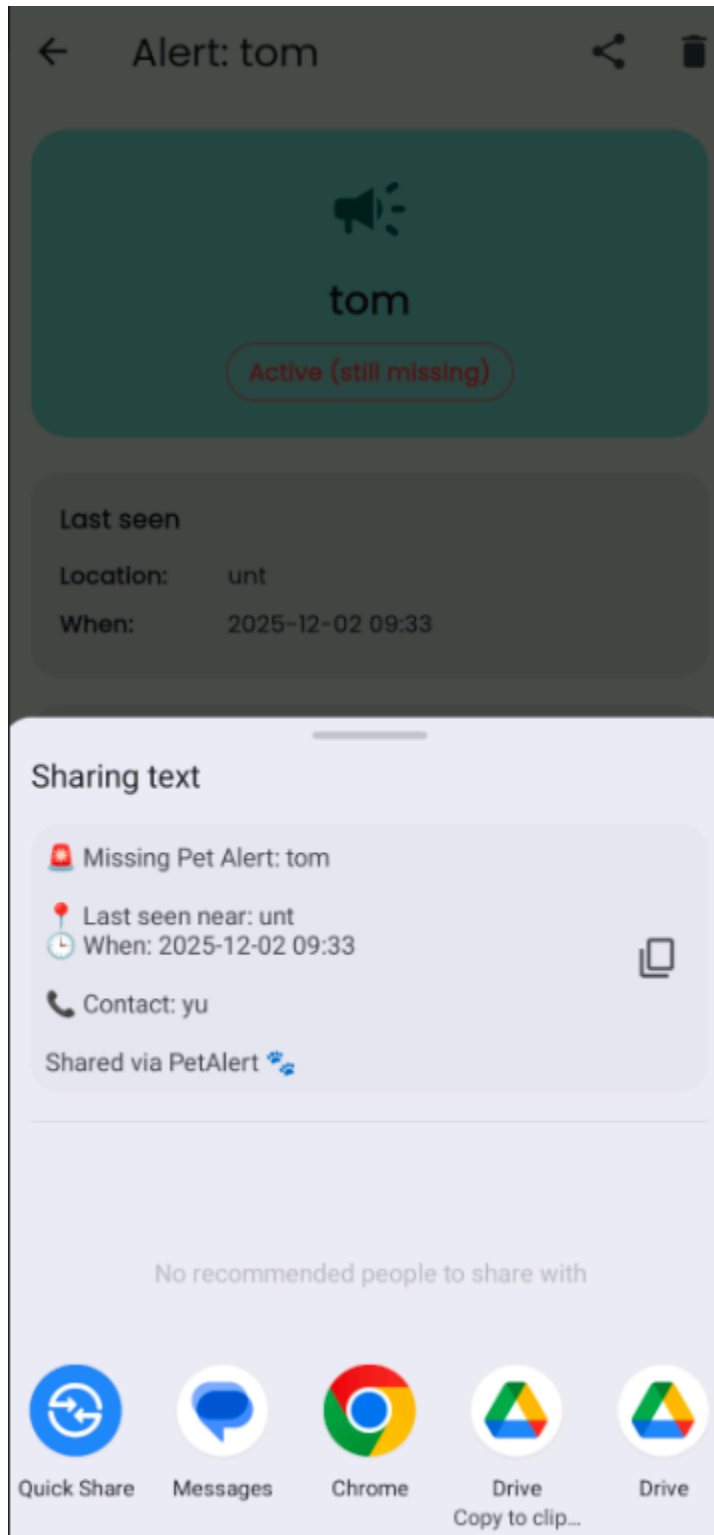
Active (still missing)

 Save Alert

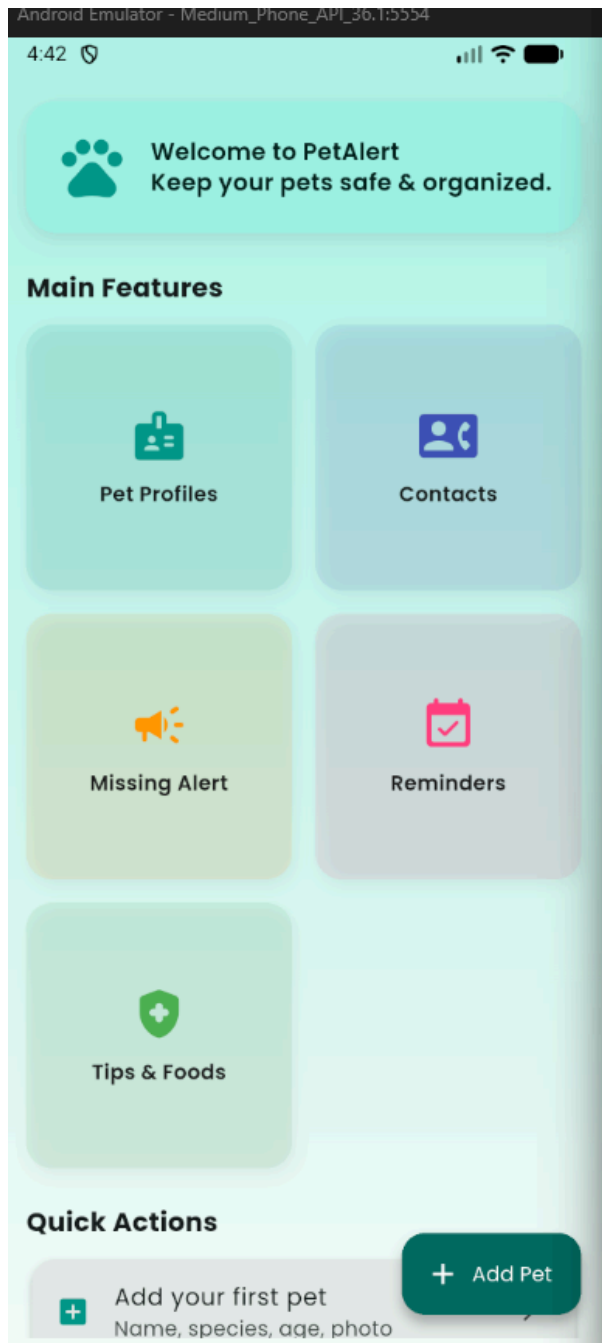
Missing Alert List



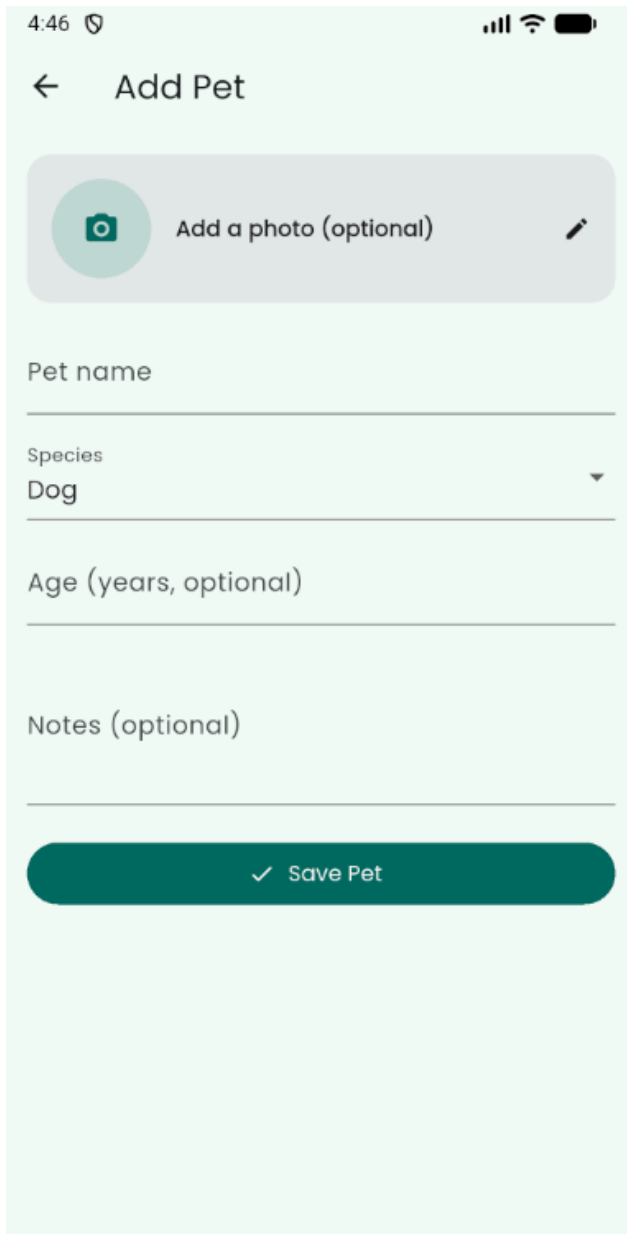
Sharing Missing Alert



Homescreen is loaded in the android emulator.



Add pet feature:



A mobile application interface for adding a new pet. The screen has a light green background. At the top, the status bar shows the time 4:46, a lock icon, and cellular/Wi-Fi/battery icons. Below the status bar is a header with a back arrow and the text 'Add Pet'. The main form area contains several input fields: a photo upload section with a camera icon and the text 'Add a photo (optional)'; a text field for 'Pet name'; a dropdown menu for 'Species' with 'Dog' selected; a text field for 'Age (years, optional)'; and a text area for 'Notes (optional)'. At the bottom is a large green button with a white checkmark and the text 'Save Pet'.

4:46

← Add Pet

Add a photo (optional)

Pet name

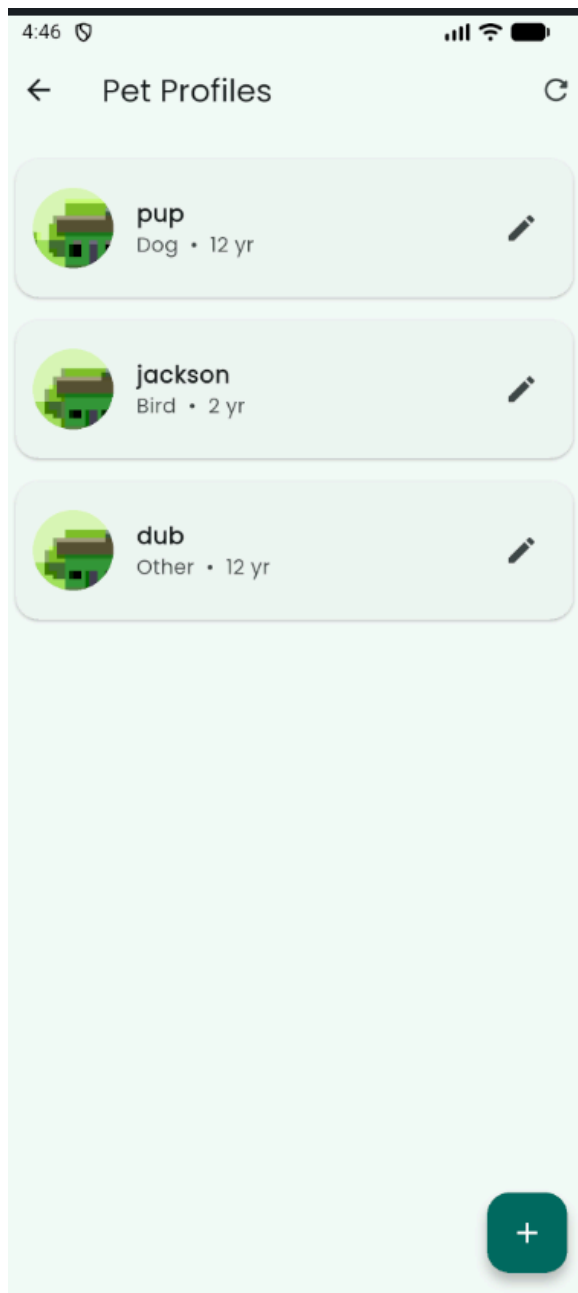
Species
Dog

Age (years, optional)

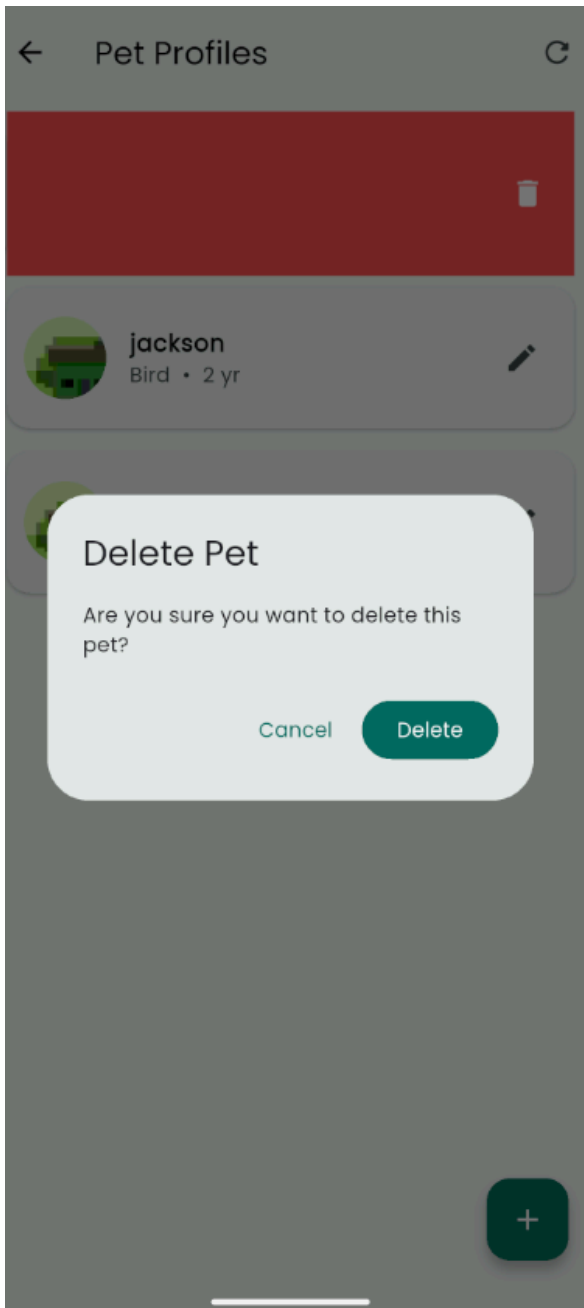
Notes (optional)

✓ Save Pet

Pet Profiles feature:



Delete profile feature



We added the microchip option in missing alert screen

Android Emulator - Medium_Phone_API_36.1:5554

11:03

← Create Missing Alert

Last seen location

Last seen date & time
Not set

Contact name (optional)

Contact phone (optional)

Notes (optional)

Optional details

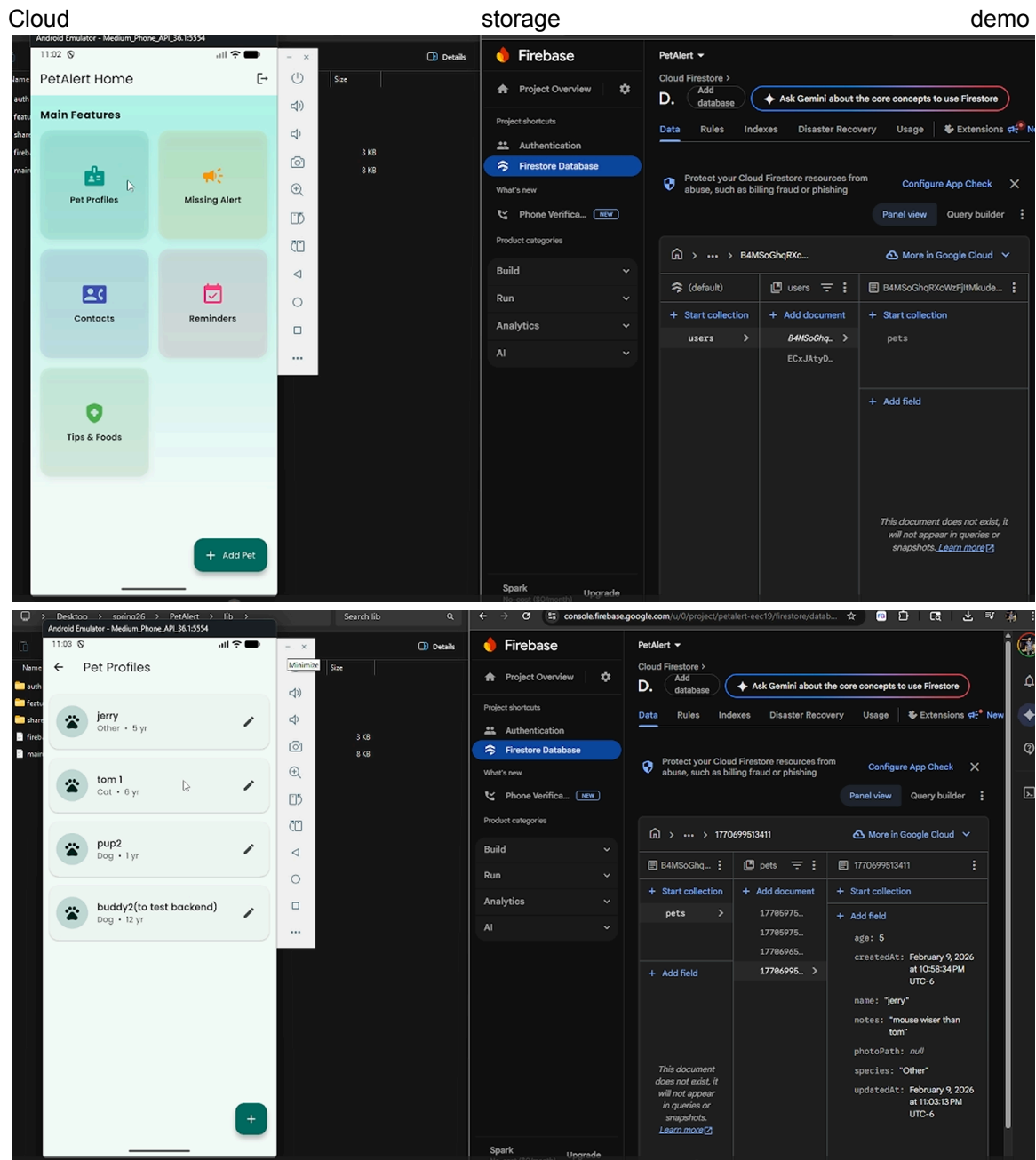
Microchip ID, distinguishing marks

Microchip ID (optional)
e.g., 9851210XXXXXX

Distinguishing marks (optional)

Status
Active (still missing)

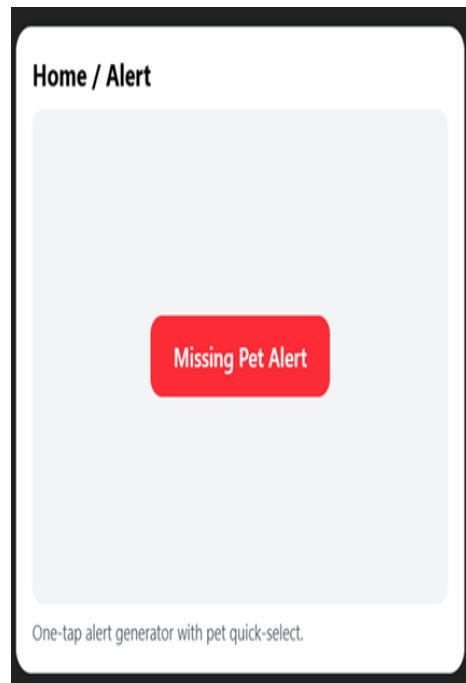
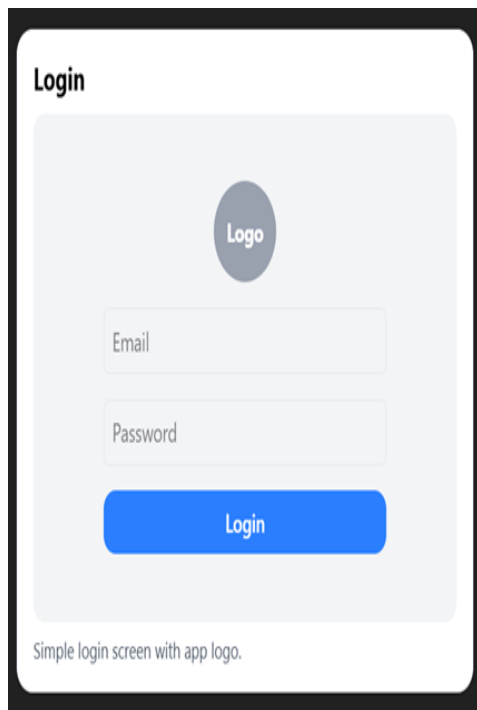
✓ Save Alert



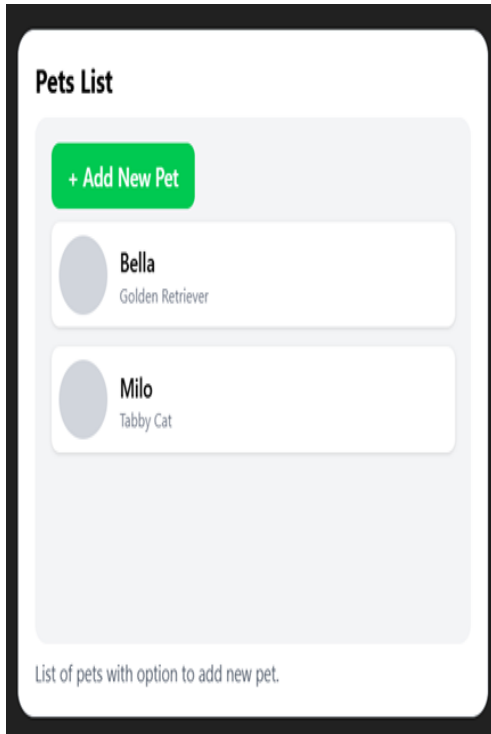
User Interface Wireframe(s)

This is the login page where the user logs in to the PetAlert app

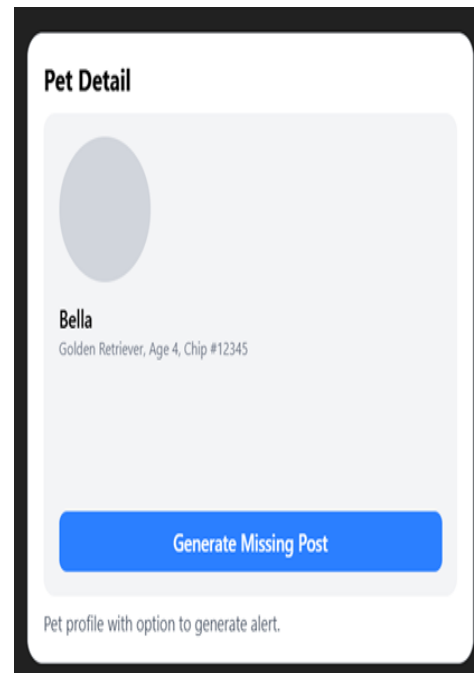
This is a missing pet alert page, where a user needs to report a missing pet by clicking the button



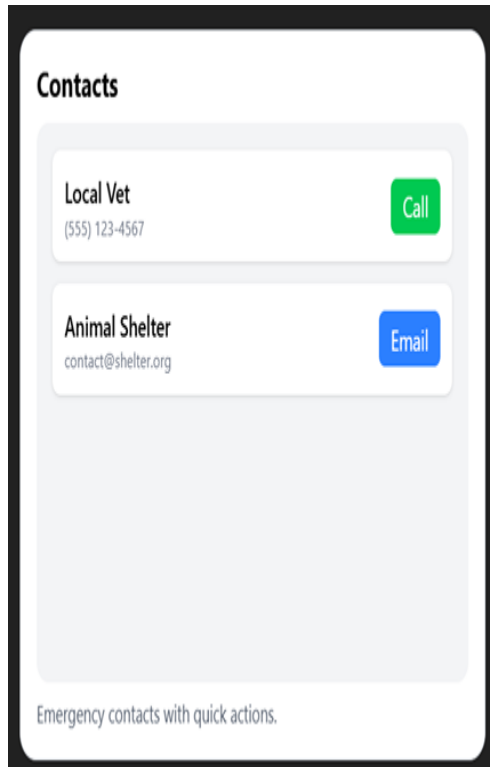
This displays a list of pets, that the user inputs. If the user wants to add a new pet there is a green button above.



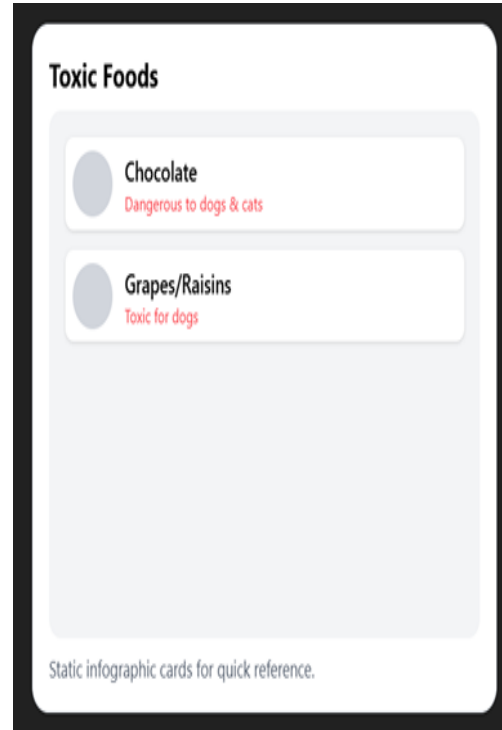
This is a pet profile page where it shows the pet account info



This page lists all the emergency contacts for that specific pet in case they need help



This page will show the infographic of toxic foods that pets can't eat



Design Summary

The design of PetAlert advanced further in Sprint 4 with the migration from local storage to a cloud-integrated architecture using Firebase Firestore and Firebase Storage. While Sprint 3 introduced authentication, Sprint 4 ensured that authenticated users now have synchronized cloud-based pet data accessible across devices.

Pet profiles, including newly added microchip number fields and uploaded images, are now stored securely in Firestore and Firebase Storage. This improves reliability, persistence after app restarts, and cross-device access while maintaining the clean and accessible interface introduced in earlier sprints.

UI refinements were preserved to maintain consistency, while backend services were updated to support asynchronous Firestore data retrieval. The design continues to

prioritize clarity, minimal navigation depth, and fast emergency access through the Missing Pet Alert module.

These updates transition PetAlert from a hybrid local prototype into a scalable cloud-ready system suitable for real-world deployment.

Design Rationale

The design decisions in Sprint 4 focused on scalability, persistence, and production readiness. Migrating pet data from local storage to Firebase Firestore ensures that users do not lose information if they switch devices or reinstall the app. Firebase Storage was integrated to manage image uploads efficiently and securely.

Adding the microchip number field enhances the completeness of pet identification data, aligning the application more closely with real-world pet recovery requirements.

The interaction flow remains intentionally simple: add or edit pet → store securely in Firestore → retrieve dynamically when needed. The architecture now supports real-time cloud synchronization while preserving usability and performance.

These changes strengthen the application's backend foundation without increasing UI complexity, maintaining the emergency-first philosophy established in earlier sprints.